[INCH-POUND]
A-A-3047
November 6, 1996

COMMERCIAL ITEM DESCRIPTION

CUTTER, BOLT

The General Services Administration has authorized the use of this Commercial Item Description for the procurement of the type II, class 1, 2, 4, and 6 cutters specified in GGG-C-740D for all federal agencies.

1. SCOPE.

1.1 Scope. This Commercial Item Description covers rigid-head adjustable bolt cutters generally used in cutting bolts, rivets, and steel rods.

2. CLASSIFICATION.

2.1 Classification. Rigid-head adjustable bolt cutters shall be of the following classes:

Class 1	-	Angular cut (flush cutting)
Class 2	-	Center cut (round edges)
Class 3	-	Clipper cut (close cutting)
Class 4	-	Notched shear cut

3. SALIENT CHARACTERISTICS.

3.1 Design. Rigid-head bolt cutters shall consist of a cutter head, handles, a means of adjusting, and bumpers or stops. All parts having the same manufacturer's part number shall be capable of being completely interchangeable, replaced, or adjusted without requiring any modification. With the exception of the various jaw configurations, all cutters shall be similar to Figure 1.

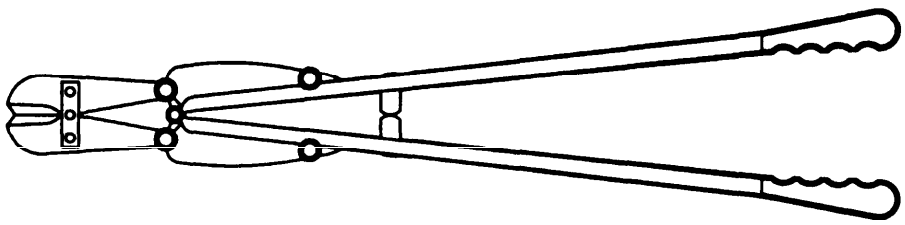


Figure 1. Handles, adjusting, levers, and bumpers for rigid-head, adjustable, bolt cutter.

3.1.1 Cutter head. The cutter head shall consist essentially of a pair of jaws, two jaw straps, and two strap bolts.

3.1.2 Jaws. The jaws shall be of high-grade tool steel, of uniform quality, smoothly finished, and free of hard spots, cracks, or other defects. The thickness of each jaw shall be not less than 85 percent of the diameter of the rod capacity. The difference in the thickness of each jaw at any point shall be negligible. The two jaws shall be

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pivotally mounted and shall have a suitable means provided for equalizing the movement of the jaws to prevent sideplay or misalignment. Additional sets of jaws may be ordered when required (see 7.2). The head and jaw configurations and hardness requirements are defined as follows:

3.1.2.1 Class 1 - Angular cut (flush cutting). The angular cut (flush cutting) bolt cutter shall have jaws with an angle of not less than 25 nor more than 35 degrees from the base plane of the jaw shanks. The obtuse angle of the jaws shall be clear and not bind with the jaw straps in any open position. The hardness of the jaws shall be between 53 and 58 inclusive on the Rockwell C scale when taken on the bevel approximately 1/8 inch from the cutting edges. The configuration of the jaws shall be similar to Figure 2, and shall conform to the length, cutting capacity, and weight requirements specified in Table I (see 7.2)

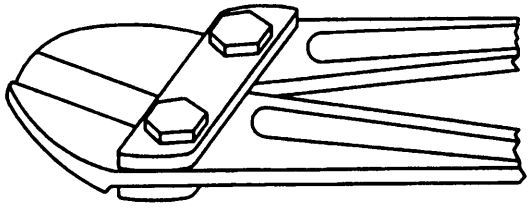


Figure 2. Cutting head for class 1, rigid-head bolt cutter, adjustable, angular cut (flush cutting).

TABLE I. Class 1, rigid-head bolt cutter, adjustable, angular cut (flush cutting)

Length +3.0/-1.0 (Inches)	Cutting capacity (Round mild steel rod) (Inch)	Weight, maximum (Pounds)
14	1/4	3-3/4
24	3/8	7-1/2
36	9/16	15

3.1.2.2 Class 2 - Center cut (round edges). The center cut (round edges) bolt cutter shall be suitable for cutting reinforcing rods and similar hard materials commonly used in general construction and wrecking work. The hardness of the jaws shall be between 57 and 62 inclusive on the Rockwell C scale when taken on the bevel approximately 1/8 inch from the cutting edges. The cutter shall be similar to Figure 3, and shall conform to the length, cutting capacity, and weight requirements of Table II (see 7.2).

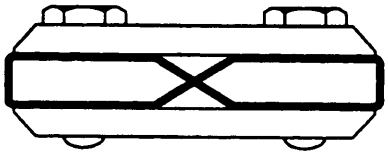


Figure 3. End view of jaws of class 2, rigid-bolt cutter, adjustable, center cut (round edges).

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TABLE II. Class 2, rigid-head bolt cutter, adjustable, center cut (round edges)

Length +3.0/-1.0 (Inches)	Cutting capacity (Round concrete-reinforcing rod) (Inch)	Weight, maximum (Pounds)
14	3/16	3-3/4
18	1/4	5
24	5/16	7-1/2
30	3/8	10-1/2
36	7/16	15
42	1/2	20

3.1.2.3 Class 3 - Clipper cut (close cutting). The clipper cut bolt cutter shall be suitable for close cutting of bolts, screws, rods and wire. The hardness of the jaws shall be 50 minimum on the Rockwell C scale when taken on the bevel approximately 1/8 inch from the cutting edges. The cutter shall be similar to Figure 4 and shall conform to the length, cutting capacity, and weight requirements of Table III (see 7.2) .

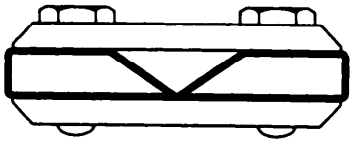


Figure 4. End view of jaws of class 3, rigid-head bolt cutter, adjustable, clipper cut (close cutting).

TABLE III. Class 3, rigid-head bolt cutter, adjustable, clipper cut (close cutting)

Length +3.0/-1.0 (Inches)	Cutting capacity (Round mild steel rod) (Inch)	Weight, maximum (Pounds)
14	1/4	3-3/4
18	5/16	5
24	3/8	7-1/2
30	1/2	10-1/2
36	9/16	15
42	5/8	20

3.1.2.4 Class 4 - Notched shear cut. The notched shear cut bolt cutter shall be suitable for cutting wire rope and cable having a minimum hardness of Brinnel 400 (Rockwell C-42). The jaws shall be notched so as to lock the cable in place during the cut and to minimize crushing or deforming of the cable. The cutting edges of the jaws, including the notches, shall pass each other in such a manner as to complete the shear cut. The hardness of the jaws shall be between 60 and 63 inclusive on the Rockwell C scale when taken 1/8 inch from the cutting edges. The jaw

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configuration shall be similar to Figure 5, and the cutter shall conform to the length, cutting capacity, and weight requirements of Table IV (see 7.2).

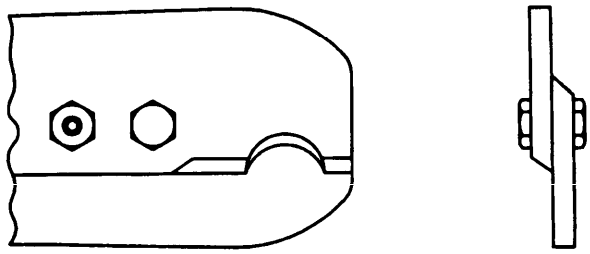


Figure 5. End of cutting head for class 4, rigid-head bolt cutter, adjustable, notched, shear cut.

TABLE IV. Class 4, rigid-head bolt cutter, adjustable, notched, shear cut

Length ± 1.0 (Inches)	Cutting capacity (Hard wire rope and cable) (Inch)	Weight, maximum (Pounds)
14	1/4	3-1/2
24	3/8	7
36	5/8	15
42	3/4	18

3.1.3 Jaw straps. The jaw straps shall be of steel and may be either the integral bolt type or the treaded type. If of the integral bolt type, the jaw bolts shall be pressed into one strap and secured to prevent rotation of the bolts, and each of the bolts shall be provided with positive locknuts for securing the jaws. If of the threaded type, the strap on the base plane of the jaws shall be threaded to properly accommodate the strap bolts for securing the jaws. Lock plates, lockwashers, or other positive means shall be provided to prevent the bolts from becoming loose or rotating. The protruding portion of the bolts shall be held to a minimum to prevent interference in confined areas.

3.1.4 Handles. The handles shall consist of two lever arms made of high grade malleable iron castings steel castings, steel forgings, or steel tubing. Tubular handles shall be provided with cushion-type hand-grips at least 3-1/2 inches in length. The lever arms shall have either toggle or geared action.

3.1.5 Adjusting means. The adjusting means shall be high grade malleable iron castings, steel, steel castings or steel forgings. An adjusting means shall be provided with a threaded component for adjustment so that the distance between the holes for the tail end of the jaws may be increased 1/4-inch minimum, or the distance between cutter jaws decreased 5/64 inch to compensate for wear of the cutting jaws.

3.1.6 Bolts. The bolts shall be of hardened steel and shall have a hardness of between 30 and 40 inclusive on the Rockwell C scale. The bolts shall be case hardened and have a hardness of between 80 and 90 inclusive on the Rockwell 15 N scale (superficial). The diameter of the bolts shall be not less than the thickness of the jaws, and the thickness of the bolt heads shall not exceed 50 percent of the bolt diameter. The threaded portion of the bolt shall not extend more than one full thread into the jaw bearing surfaces.

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3.1.7 Bumpers or stops. All bolt cutters, except the class 2, 42-inch size, shall have either adjustable lock screws or high grade rubber bumpers. The rubber bumpers shall be capable of withstanding compression to at least one-half of their original length without indication of failure. The outer faces of the rubber bumpers shall be protected by metal washers or rivet heads of a diameter that is equal to the diameter of the rubber bumpers when not compressed. The adjustable lockscrew bumpers shall be of steel and properly threaded to allow adjustment of the handle and jaw closure. The bumpers or stops shall be located and so arranged on the handles as to preclude the possibility of the handles being closed beyond the effective cutting point and to eliminate pinching the hands of the operator. The class 2, 42-inch size cutter, and cutters with tubular handles, shall be constructed with solid stops which have been integrated into the handles.

3.2 Identification marking. The tool shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture, unless otherwise specified. All identification markings shall be engraved, etched, molded, or indented directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item.

3.3 Workmanship. Details of workmanship shall be in accordance with the best commercial practice. Paints, coatings, platings, and finishes shall be smooth, adherent, continuous, and not stained or discolored. External surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be free from manufacturing workmanship defects (e.g., sharp or rough external edges, corners, or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified (using conversion tables contained in the latest revision of Federal Standard 376) and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

5. QUALITY ASSURANCE PROVISIONS.

5.1 Product conformance. The product provided to the Government shall conform to the contractually specified requirements. For product characteristics and requirements which are not contractually specified, the provided product shall conform to the producer's own drawings, standards, specifications, and quality assurance practices for the highest quality product commercially offered by the producer. If the product is not commercially offered by the producer, the best commercial practices shall apply. The Government reserves the right to require proof of such conformances prior to first delivery and thereafter as provided for under the provisions of the contract.

5.2 Responsibility for inspection. Unless otherwise specified, the contractor is responsible for the performance of all inspection requirements and may use any commercial facilities (including the contractor's own facilities) suitable for performance of the inspection requirements, unless disapproved by the Government. The Government reserves the right to perform any inspections deemed necessary to assure the item conforms to the specified requirements.

5.3 Bid sample(s). When a bid sample requirement is specified in the solicitation, the bid sample(s) shall be inspected for all salient characteristics by the Government. A failure of any bid sample to meet the salient characteristics shall be cause for rejection.

5.4 First article sample(s). When a first article requirement is specified in the contract, the first article sample(s) shall be inspected for all salient characteristics. A failure of any first article sample to meet the salient characteristics shall be cause for rejection.

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5.5 Hardness test. Portions of sample bolt cutters for which applicable hardness requirements have been specified shall be tested in accordance with ASTM E18. The readings shall conform to the hardness as specified in 3.1.2, 3.1.2.1, 3.1.2.2, 3.1.2.3, 3.1.2.4, and 3.1.6. Grinding of specimens or other preparations for hardness readings shall be done in such a manner as to avoid tempering or distortion. Failure to meet any specified requirement shall be cause for rejection of the bolt cutter and the lot it represents.

5.6 Cutting test. Each sample bolt cutter shall cut a minimum of 10 pieces of steel of the maximum diameter specified in the applicable tables under "cutting capacity" for each cutter. The cuts shall be made as close as possible to the outermost point of the cutting edge of the jaw. No assistance such as bending, pulling, or twisting of the steel by the operator shall be allowed. At the completion of this test, visual examination of the cutters shall show no evidence of flaws, cracks, undue wear, turning, or breaking out of any part of the cutting edge of the jaw or permanent distortion of any part. Failure of any sample to meet this test shall be cause for rejection of the cutter and the lot it represents.

5.7 Jaw strength test. Each sample bolt cutter, with the exception of the class 4 cutter, shall be subjected to the jaw strength test in accordance with the requirements of Table V. The test shall be conducted with flat strip low-carbon steel having a Brinell hardness of 110 to 140 (comparable to approximately 64 to 76 on the Rockwell B scale) and of a size specified for each nominal length of cutter. The steel shall be placed between the cutting jaws at the outermost end and a force or pressure load sufficiently applied to the handles to bring the bumpers together. The force or pressure load shall be released and the jaws shall return to their normal position. The cutters shall be examined and no failure, perceptible deformation, or damage to the cutting jaws, jaw bolt, or straps shall have occurred as a result of this test. Failure of the sample to meet this test shall be cause for rejection of the cutter and the lot it represents.

TABLE V. Jaw strength test

	Flat strip steel	
Nominal length of cutter (Inches)	Thickness (Inch)	Width (Inch)
14	1/16	1/2
18	1/16	1/2
24	3/32	1/2
30	1/8	1/2
36	3/16	1/2
42	3/16	1/2

5.8 Maximum cutting load test. Class 1, 2, and 3 bolt cutters shall be subjected to a maximum cutting load test in accordance with Table VI. The test shall be conducted by placing the test piece into the jaws of the cutter a distance of 1/2 inch from the nose of the jaws. Two preliminary cuts shall be made in the test piece before any cutting load determinations are made. The purpose of the preliminary cuts is to "settle" the tool. Three cuts shall then be made and the cutting load determinations shall be averaged. The average cutting load shall not exceed the maximum load specified in Table VI, as applicable. After the test, the cutters shall be examined and no failure, perceptible deformation, or other impairment shall have occurred as a result of the test. Failure of the sample to meet the load cutting test shall be cause for rejection of the cutter and the lot it represents.

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TABLE VI. Maximum cutting load

Type	Class	Size (Inches)	Test rod (Inch)	Maximum cutting load (Pounds)
			1018 CRS 1/	
II	1	14	1/4	120
II	1	24	3/8	145
II	1	36	9/16	230
			Reinforcing rod 2/	
II	2	14	3/16	60
II	2	18	1/4	90
II	2	24	5/16	90
II	2	30	3/8	125
II	2	36	7/16	125
II	2	42	1/2	185
			1018 CRS 1/	
II	3 & 4	14	1/4	120
II	3 & 4	18	5/16	145
II	3 & 4	24	3/8	135
II	3 & 4	30	1/2	185
II	3 & 4	36	9/16	210
II	3 & 4	42	5/8	230

1/ AISI-1018 - Cold drawn steel bars, for testing classes 1, 3, and 4 bolt cutters.
2/ Intermediate grade reinforcing bar, new billet, for testing class 2 bolt cutters.

5.9 Rubber bumper compression test. Sample bolt cutters having rubber bumpers shall be subjected to a compression test. The method of determining the amount of compression of the bumper may be obtained by measuring, with the use of calipers, the overall length of the bumper before and during the compression load. The bumper shall withstand compression to at least one-half of the original length without indication of failure. Failure to meet the requirement of this test shall be cause for rejection of the bolt cutter and the lot it represents.

6. PACKAGING.

6.1 Packaging. Requirements of preservation, packing, packaging, and marking of packages shall be as specified in the contract or order.

7. NOTES.

(This section contains information of a general or explanatory nature that is not mandatory.)

7.1 Addresses for obtaining copies of referenced documents:

Federal Specifications and Standards: Federal Supply Service Bureau, Specification Section, Suite 8100, 470 L'Enfant Plaza, SW, Washington, DC 20407.

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Federal Acquisition Regulations (FAR): Government Printing Office, Superintendent of Documents, Washington, DC 20402-9371.

ANSI/ASQC Z1.4: American Society for Quality Control, P.O. Box 3005, 611 East Wisconsin Avenue, Milwaukee, WI 43201-4606.

ASTM E18: American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

AISI 1018: American National Standards Institute, 1430 Broadway, New York City, NY 10018.

7.2 Ordering data. Purchasers should select the preferred options permitted herein and should include the following information in procurement documents:

(a) Title, number, and date of this Commercial Item Description.
(b) When applicable, the appropriate NSN(s).
(c) When applicable, the specific classes, as specified in the Commercial Item Description.
(d) If bid samples are required, the number of samples required, and the subjective inspection requirements.
(e) If first article samples are required and the number of samples required.
(f) If lot sampling inspection is required and the applicable lot sampling requirements. For example:

Sampling for inspection. Sampling for inspection (both examinations and testing) shall be in accordance with ANSI/ASQC Z1.4.

Inspection. Each sample tool shall be examined and tested for all salient characteristics. The Inspection Level shall be S-2 with an Acceptable Quality Level of 4.0, expressed in terms of defects per hundred units.

(g) Additional sets of replacement jaws, if required (see 3.1.2).
(h) Length, cutting capacity, and weight required (see 3.1.2.1, 3.1.2.2, 3.1.2.3, 3.1.2.4).
(i) What preservation, packing, packaging, and marking of packages are required.

7.3 National Stock Numbers (NSNs). The following list of NSNs corresponds to the classes specified in this CID. This list may not be indicative of all possible NSNs associated with this CID.

NSN	LENGTH	CAPACITY
Class 1 Bolt Cutter		
5110-00-224-7055	14 inch	1/4 inch
5110-01-380-5089	Replacement jaws for 5110-00-224-7055	
5110-00-224-7056	24 inch	3/8 inch
5110-01-381-0790	Replacement jaws for 5110-00-224-7056	
5110-00-224-7057	36 inch	9/16 inch
5110-00-380-5088	Replacement jaws for 5110-01-380-5088	
Class 2 Bolt Cutter		
5110-00-224-7064	30 inch	3/8 inch

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Class 3 Bolt Cutter		
5110-00-596-9162	18 inch	5/16 inch
5110-01-378-4608	Replacement jaws for 5110-00-596-9162	
5110-00-596-9156	24 inch	3/8 inch
5110-01-380-5091	Replacement jaws for 5110-00-596-9156	
5110-00-596-9155	30 inch	1/2 inch
5110-01-380-5095	Replacement jaws for 5110-00-596-9155	
5110-00-188-2524	36 inch	9/16 inch
5110-01-381-0808	Replacement jaws for 5110-00-188-2524	
Class 4 Bolt Cutter		
5110-00-224-7053	14 inch	3/4 inch

MILITARY INTERESTS:

NONE: DoD has no registered interest in revisions or notices to this Commercial Item Description until further notice.

PREPARING ACTIVITY:

GSA - FSS

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